

CORE — MASTER SPECIFICATION — FULL DETAIL

Version 3.6 • 21 August 2026

Technical • Functional • Architectural • API • Security • Database • Administration • Governance

Document role: Primary CORE master reference for architecture, development handover and governance.

Architecture status: Shared foundation; application-specific business logic remains outside CORE.

v3.6 scope: Updates the previous v3.5 specification with the verified post-v3.5 CORE work: Universal Application PRE_LAUNCH/PUBLIC, Notifications & User Engagement, Dashboard Actions & Resource References, Premium-Locked Content, Universal Affiliate, and Purchasable Points.

Critical boundary: PRE_LAUNCH/PUBLIC applies to CORE-connected applications, not to the CORE application itself.

CURRENT IMPLEMENTATION STATUS

Area	Current status
PRE_LAUNCH / PUBLIC	Implemented for all CORE-connected applications; CORE itself excluded.
Notifications & Engagement	Implemented; shared pipeline, preferences, deduplication, activity/inactivity support.
Dashboard Actions / Resources	Implemented; generic cross-app actions and resource references.
Premium-locked content	Implemented; protected values resolved server-side.
Affiliate	Implemented; Admin-approved offers, percentage/fixed commission, attribution, commission ledger and payout preparation.
Purchasable Points	Implemented; extends the existing Rewards ledger, no second Points balance.
External payout transfer	Not fully automated; payout sweep prepares eligible payouts, but no external payout provider is configured.
Known pre-existing TS errors	Four unrelated pre-existing errors remain; no new errors reported by the latest task.

1. Core Architecture & Boundary

- CORE is the central shared SaaS platform and operating layer for current and future applications.
- CORE owns shared identity, users/profiles, applications/capabilities, roles, Premium/entitlements, billing, Rewards/Event Engine, advertising, offers, notifications, messaging, security, audit, API, shared UI and localization.
- Applications retain their own business databases and application-specific business logic.
- CORE must remain application-agnostic. Shared code must not branch on application name, slug or application-specific business concepts.
- Application products such as BosniaFans Shop products remain application-owned even when CORE provides shared commercial or Affiliate infrastructure.

2. Non-Negotiable Development Rules

- Single Source of Truth.
- Reuse Before Create.
- Configuration First.
- Server-Side Authority for prices, permissions, rewards, entitlements and payment state.
- Security by enforcement; UI visibility is never an authorization boundary.
- Do not create duplicate Identity, Premium, Rewards, Notifications, Dashboard Actions or Affiliate systems.
- Production database changes use new migrations, live verification and documentation.

3. Identity, Users, Applications & Capabilities

- One shared CORE identity is used across consuming applications.
- Applications are registered once and participate through app_id, application context and capabilities.
- Capability checks that control access or behavior are evaluated server-side.
- Application-specific data remains isolated.
- CORE Admin authorization remains centrally enforced; no duplicate per-app administrator system is introduced unless explicitly required by architecture.

4. Premium, Subscriptions & Entitlements

- Global Premium is user-level and shared across CORE-connected applications.
- Application-specific commercial products and entitlements remain separate from Global Premium.
- Premium/entitlement decisions are server-side.
- CORE provides shared Premium resolution and entitlement infrastructure while applications define their own Premium-specific business benefits.
- Premium pricing and product configuration remain centrally configurable and are not hardcoded in application code.

5. Billing & Payment Architecture

- Stripe is the established production payment rail. PayPal functionality exists subject to the previously documented empirical verification status.
- Payments, signed references, webhook verification, refunds and payment auditability remain centralized.
- Client-supplied amount, product, entitlement or payment-success claims are never trusted.
- Verified payment state is required before granting subscriptions, entitlements or purchased Points.

6. Universal Rewards / Event Engine

- CORE has one universal Rewards/Event Engine.
- It provides Points balance, reward ledger, reward events/actions, limits, milestones, reward fulfillment and Reward Boosts.
- No second Rewards or Points engine may be introduced.
- Applications emit generic meaningful events; application business logic remains outside CORE.

7. Dashboard — Control Center

- The Dashboard is the shared authenticated control center.
- It prioritizes identity, activity, rewards and user-benefit actions before paid advertising.
- It can expose My Applications, user resources, For You, Complete, New, Discover, notifications, Premium state, activity and other applicable CORE surfaces.
- Existing Dashboard Offers remain the commercial offer mechanism. Dashboard Actions are the generic action/prompt mechanism.

8. Notifications & User Engagement

- CORE provides one shared notification pipeline for in-app and email-capable communication.
- Notification categories, read state, deduplication, delivery status/error state and user notification preferences are part of the shared system.
- Preference fields are enforced by the shared send path.
- Meaningful triggers include messages, Premium milestones, important application events, achievements and Admin-published offers/actions where configured.
- Inactivity reminders are supported through the shared activity/inactivity mechanism. This is an engagement mechanism, not a page-view surveillance system.
- Email delivery depends on external provider configuration; in-app notification behavior remains available when email credentials are not configured.
- Applications must reuse the shared notification pipeline.

9. Central User Activity

- CORE provides meaningful cross-app user activity through the existing Event Engine/activity aggregation.
- Activity does not mean logging every click or page view.
- Users see their own relevant activity. Admin/User 360° may see authorized cross-app activity.
- Applications may emit generic meaningful activity events without moving application business data into CORE.

10. Dashboard Actions & Resource References

- Dashboard Actions are generic, non-commercial or action-oriented prompts separate from commercial Dashboard Offers.
- Admin can create/configure actions with title, description, CTA, destination, application context, active state, ordering and optional validity.
- Applications can provide generic actions without putting application business logic into CORE.
- Resource References allow CORE to represent a user-owned external resource such as Shop, Business or Event without storing that resource's business data.
- Examples include 'Otvori svoj Shop', 'Reklamiraj svoj Business', 'Complete your profile' and 'Postani Affiliate'.
- CORE does not hardcode BosniaFans, Eshop.ba or other applications into the generic mechanism.

11. Premium-Locked Content

- CORE provides a reusable server-side content-lock primitive based on Premium/Entitlement eligibility.
- Protected values are never sent to a Standard viewer merely to hide them with CSS/JavaScript.
- Standard users may receive existence/locked state; eligible Premium users receive the protected value.
- The existing Global Premium and Entitlement systems determine eligibility.
- Revoked Premium access takes effect on the next authoritative fetch; client-side caching is not an authorization mechanism.

12. Universal Application PRE_LAUNCH / PUBLIC Standard

- This standard applies to every CORE-connected application and explicitly does not apply to CORE itself.
- Every new CORE-connected application starts in PRE_LAUNCH.
- PRE_LAUNCH uses the same final/main production domain. It is not a separate staging domain.

- While PRE_LAUNCH, '/' is the public Pre-Launch Front Page. It can contain application logo, main image/banner, application-specific information, development/preparation message, social profiles and optional contact information.
- Normal public visitors may access only the Front Page. Direct access to protected routes must be blocked at the authoritative access-control boundary.
- Application Admin has full access during PRE_LAUNCH.
- Admin may authorize selected application-scoped test users.
- Only an explicit Admin transition changes PRE_LAUNCH to PUBLIC.
- PUBLIC transition does not change domain, URLs, routes, links or architecture.
- The mechanism is generic and inherited by all current and future CORE-connected applications.
- CORE infrastructure endpoints that legitimately require public access are not indiscriminately blocked by the application gate.

13. Universal Affiliate System

- CORE provides one central Affiliate system for CORE products and products/services owned by CORE-connected applications.
- Affiliate can be globally enabled/disabled and application capability controlled. Individual Affiliate offers are OFF by default until Admin explicitly enables them.
- Admin decides which concrete products/offers may be promoted and sets the commission.
- Commission supports percentage or fixed monetary amounts.
- Examples: CORE Premium 20%; BosniaFans Sneakers 10%; Advertising package €15 fixed. These are examples, not hardcoded values.
- Each offer identifies its source as CORE or an application. The originating application remains owner of product, order, payment, fulfillment, refund and business rules.
- Affiliate user statuses include NOT_AFFILIATE, ACTIVE, SUSPENDED and DISABLED.
- Active users receive an Affiliate catalog containing only currently eligible offers.
- Referral links use non-guessable identifiers. Attribution uses a configurable attribution window and last valid Affiliate click.
- Self-referral is blocked where the originating user identity is reliably known. Duplicate transaction processing is blocked by idempotent transaction references.
- Commission is server-calculated and snapshotted at conversion time so later Admin rate changes do not change historical conversions.
- Commission lifecycle: PENDING → APPROVED → PAID, with reversal handling for refunds/chargebacks.
- Refund/chargeback reversals preserve financial history. A paid commission is not silently mutated; a reversal/negative ledger entry is used where required.
- Approved commissions below the minimum payout threshold carry forward.
- Eligible approved balances are collected by the monthly payout sweep for the next payout cycle.
- The current implementation prepares payout records automatically. Actual external transfer is not fully automated because an external payout provider is not configured.
- Affiliate does not include multilevel/network marketing, Points-to-cash conversion or a banking wallet.

14. Purchasable Points — Rewards Extension

- Purchasable Points extend the existing CORE Rewards ledger; they do not create a second balance.
- Points retain an origin distinction such as earned versus points_purchase.
- Admin can enable/disable Points purchases and configure active packages.
- Packages can define name, price, currency, Points amount, optional bonus, active state and ordering.
- Purchase flow uses the existing payment architecture and grants Points only after verified payment.
- The client cannot determine package price, Points quantity or granted balance.
- Purchased Points use the same Rewards/Premium redemption rules as the existing Points balance where those rules permit it.
- Example: if Premium requires 2,000 Points and the user has 1,250, Dashboard can show 'Još 750 Points do Premiuma' and, when enabled, offer 'Buy Points'.
- Points are not redeemable for cash and CORE does not become a wallet.

- Refund/reversal behavior preserves ledger integrity and does not silently delete financial history.

15. Affiliate + Commercial Product Boundary

- Affiliate is an overlay on existing commercial products, not a replacement for product/order/payment systems.
- CORE products such as Premium or advertising packages may be Affiliate-eligible.
- Application products such as BosniaFans Shop Sneakers or Eshop.ba products may be Affiliate-eligible.
- Admin controls eligibility and commission for each concrete offer.
- CORE manages tracking, attribution and commission; the originating application remains responsible for its own business product.

16. Advertising, Offers & Coupons

- Advertising remains a separate central commercial system with campaigns, placements, targeting, pricing, impressions, fair rotation and lifecycle states.
- Dashboard advertising remains visually secondary to user-owned Dashboard content.
- Dashboard Offers remain targeted user benefits with server-derived pricing/discount behavior.
- Public Coupons remain distinct from personalized Dashboard Offers and use their supported registration/payment redemption flow.
- Affiliate does not replace Offers, Advertising or Coupons.

17. Messaging

- CORE supports shared 1:1 messaging with canonical user pairs, read state and application provenance where applicable.
- Notifications may be generated from message events through the shared notification pipeline.

18. Administration / User 360°

- Admin/User 360° remains the central administrative surface.
- It covers users, applications, Premium, entitlements, Rewards, activity, payments, advertising, offers, communication and platform configuration.
- Affiliate administration includes global configuration, eligible offers, Affiliate users, conversions and payout records.
- Points administration includes Points packages and purchase/ledger visibility.

19. Security Architecture

- Authentication, authorization, JWT identity, protected routes, JWKS, middleware, CORS, security headers, CSRF where applicable, RLS, server validation, signed payment references, audit logs and service-role isolation remain required.
- Financial values, commission values, Points grants and entitlement state are server-authoritative.
- RLS is a database enforcement boundary but does not replace correct function grants.
- Protected Premium values must be resolved server-side.
- Affiliate and Points financial data must be isolated by user/application/admin authorization.

20. API Architecture

- CORE API remains a thin, versioned interface over shared business functions.
- Existing /v1 areas remain authoritative for identity, profiles, Premium, applications, messaging, notifications, billing, security, advertising and Rewards.
- Implemented additions include authenticated resource reference handling and Affiliate conversion/reversal endpoints.
- Exact endpoint signatures must remain synchronized with API_CONTRACT.md and the actual codebase.

21. Database Architecture

- CORE stores shared platform entities; application-specific business data remains isolated.
- Protected tables require RLS and controlled server-side access.
- Historical applied migrations must never be rewritten.

- Affiliate adds dedicated affiliate configuration/offers/links/clicks/conversions/commission/payout structures.
- Purchasable Points extends the existing Rewards ledger and payment structures rather than creating a second balance.

22. Launch, Application Lifecycle & Main Domain

- New CORE-connected applications register with CORE, configure domain/branding/capabilities, initialize PRE_LAUNCH, then undergo Admin-controlled testing.
- Public launch occurs only after application completion and verification.
- The same production domain and route structure are retained through PRE_LAUNCH and PUBLIC.
- CORE itself follows its own lifecycle and is not subject to the universal application launch gate.

23. Production Readiness / Verification Rules

- Implementation status must be represented as a matrix, not a single GO/NO-GO label.
- Database changes must be live-verified for schema, RLS, grants and constraints.
- Provider behavior must not be claimed as verified without empirical provider testing.
- Known pre-existing errors must be distinguished from new errors.
- Latest Affiliate/Points implementation reports state that migrations were applied and RLS/isolation were verified. Actual external payout transfer remains provider-dependent.

24. Non-Goals / Explicit Exclusions

- CORE is not the BosniaFans content database or Shop database.
- CORE is not the Eshop.ba Shop business database.
- CORE is not the Svadba vendor database or Ticketaria event database.
- CORE is not an n8n workflow engine or social publishing platform.
- Affiliate is not multilevel marketing.
- Points are not cash and CORE is not a banking/wallet system.
- Application-specific business logic must remain outside CORE.

25. Development Workflow & Claude Code Rules

- Identify genuine cross-application need.
- Search existing CORE components/services/entities before creating anything.
- Define the CORE/Application boundary.
- Reuse existing services and entities.
- Implement the smallest correct change.
- Run typecheck/build/lint and relevant route/API checks.
- For database changes, verify migration state, apply authorized migrations, live-verify and document.
- Do not silently fix unrelated findings.
- Do not claim browser/provider/production verification that was not performed.
- Commit and push only task-related changes; report exact files, migrations, tests, commit and push state.

26. Updated Core Module Map

- 01 Dashboard / Control Center
- 02 Identity & Authentication
- 03 Users & Profiles
- 04 Applications Registry & Capabilities
- 05 Roles & Permissions
- 06 Members / Membership Directory
- 07 Premium & Subscriptions / Entitlements
- 08 Billing & Payments
- 09 Universal Rewards & Event Engine
- 10 Advertising & Campaign Distribution
- 11 Offers & Coupons
- 12 Notifications & Communication
- 13 Messaging / Conversations
- 14 Security & Audit
- 15 API / Integration Layer
- 16 Shared UI / Design System
- 17 Localization
- 18 Administration / User 360°
- 19 Cross-cutting configuration and platform services
- 20 Dashboard Actions & Resource References
- 21 Universal Affiliate
- 22 Purchasable Points / Rewards Extension

27. Current Principal Data Domains

- Identity, profiles, applications, capabilities, roles and application membership/settings.
- Subscriptions, entitlements, payments and payment references.
- Rewards ledger, reward rules, milestones and Reward Boosts.
- Notifications, activity/events and audit.
- Advertising campaigns/placements, Dashboard Offers and public coupons.
- Dashboard Actions and Resource References.
- Affiliate configuration, Affiliates, Affiliate Offers, Links, Clicks, Conversions, Commissions and Payouts.
- Points packages and purchased-Points ledger integration.

28. Current Open / Deferred Status

- Any open security finding from the current audit remains open until its required live verification is completed.
- PayPal empirical verification remains governed by the latest documented status and must not be claimed complete without a controlled provider test.
- External Affiliate payout transfer is not fully operational until a supported external payout provider is configured and verified.
- Email notification delivery requires external email-provider configuration.
- Four pre-existing unrelated TypeScript errors remain known; the latest Affiliate/Points implementation introduced no new TypeScript errors according to its final report.

29. Source Basis & Version History

v3.6 is based on the v3.5 Master Specification and the verified implementation reports for the post-v3.5 CORE tasks. The document preserves the established terminology and architecture boundary. It distinguishes implemented behavior from provider-dependent or unverified behavior.

Version	Date	Change
3.5	17 Aug 2026	Previous full-detail master specification.
3.6	21 Aug 2026	Adds verified notification/engagement, Dashboard Actions/Resources, Premium-locked content, Universal Affiliate, purch

Document control: v3.6 supersedes v3.5 as the current full-detail CORE Master Specification. Exact source signatures, database details and API schemas remain authoritative in the implementation and API_CONTRACT.md.